

Liam Shannon

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SKILLS

Languages: TypeScript, JavaScript, Python, C/C++

Tools & Frameworks: Palantir Foundry, Ontology SDK, React, Git, Docker, Linux, ROS2

Data: Python data pipelines, data cleaning & normalization, SQL-style querying/Foundry querying logic

AI-Assisted Development: Claude Code, used daily to plan, explore, and implement on production work

Other: SolidWorks, Onshape, Altium, LTspice

EXPERIENCE

Software Developer

June 2026 - September 2026

Cavtera

Ottawa, Canada

- Exposed 10,000+ hours per week of equipment idling, worth \$1,000,000+ annually in recoverable cost, by building a full-stack Equipment Management System in Palantir Foundry for a previously untracked fleet
- Increased job-site utilization of 1,000+ trucks and pieces of equipment by shipping automated alerts
- Restored live equipment telematics by fixing broken ingestion APIs and rebuilding the Python transform pipeline that parses 10,000+ records per day across a database of millions
- Modeled the business as a Foundry ontology and wrote the TypeScript edit functions behind it, giving non-technical staff a governed way to update production records
- Shipped the operator-facing front end as a React and TypeScript application on Foundry's Ontology SDK (OSDK), so site staff could act on live fleet data directly
- Reviewed code and wrote and ran unit tests across the Python and TypeScript codebase

Software Developer

May 2025 - April 2026

Thomas Cavanagh Construction Limited

Ottawa, Canada / Remote

- Digitized dispatch for a 100+ driver fleet by building custom dispatch software in Palantir Foundry
- Eliminated 4+ hours of daily dispatcher calls and dozens of spreadsheets by modeling fleet data in an ontology
- Reduced compute costs from \$100+/week to single digits by caching data locally instead of querying the ontology
- Delivered the dispatcher-facing tooling as Foundry Workshop applications backed by Python transforms

Automation Manager

July 2026 - Present

Queen's Aerospace Design Team

Kingston, Canada

- Directing a 15-person software team, delegating tasks, and onboarding new ROS 2 contributors
- Developing a software stack for SAE Aero 2027, a fixed-wing autonomous drone competition
- Managing a setup in Docker allowing cross-platform builds on macOS, Windows, and Linux of an Ubuntu 24.04 system
- Training a YOLO object-detection model with Roboflow to identify ground targets from the air
- Implementing physical AI on the airframe by integrating the vision model into the ROS 2 autonomy stack, so onboard perception drives live flight decisions

PROJECTS

Autonomous Mars Rover Robot | ROS, Raspberry Pi 4, Arduino, SLAM

January 2026 - April 2026

- Designed a fully autonomous four-wheeled rover using ROS on a Raspberry Pi 4 for decision-making, sensor processing, and simultaneous localization and mapping
- Built a low-level Arduino control system handling motor drivers, encoder feedback, and PID control for precise motion

Line Following Robot | Arduino, C++, Closed-Loop Control

January 2025 - April 2025

- Developed a line-following robot using an Arduino to program motor actions based on closed-loop sensor feedback
- Tuned algorithms based on sensor data and optimized performance by integrating hardware and software subsystems

EDUCATION

Queen's University

Expected May 2028

BASc in Mechatronics and Robotics Engineering | Dean's Scholar | GPA: 4.19/4.3

Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Robotics Design II